HEAPLINE · CALCULATION

HeapLine sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-11-03	Amish Chadha	Kit 1.9.1 upgrade: masses re-read from the D3 model (welds, fasteners and clips now modelled): heaviest item 20.9 kg, carried kit 150 kg, capstan wave 21.4 kg each; no design or target changed; requirement IDs PR-001 to PR-015 added to the requirement map
0.2	2026-10-03	Amish Chadha	Amish's requirement decisions 20A, 21A, 22A and 23C (HPL-DDR-003): 18 mm probe tip; four haulers on debris; two carrying waves; shared capstan set, reused crates and PPE from the cooperative; R9 restated
0.1	2026-10-03	Amish Chadha	First sizing note on the constructable design (HPL-DDR-002); capstan, probe, crawl board, stretcher, clamp, masses, deployment time and cost

HeapLine sizing calculations

The constructable design meets R1, R3, R4, R10 and R11 on paper. After Amish's requirement decisions of 2026-10-03 (HPL-DDR-003; Amish: "i agree with all the 46 recommendations you provided. please proceed."), R2 (probe depth) is met on paper with the 18 mm tip, R5 (casualty transport) is met under the drill-card rule of four haulers on debris, R8 (portable) is met by carrying the kit in two waves, and R9 (cost) is met as restated, with one capstan set shared by neighbouring sites. Every figure below is printed by `docs/04-calcs/sizing.py` with the tag shown in brackets, from the geometry in `cad/src/model.py` and the prices in `bom/bom.csv`. These are first-principles estimates for a paper proof of concept, not test results.

> **Safety:** The capstan stores energy in a loaded rope. Every capstan part is sized on an 8 kN design tension, above the top of the shear pin's release band, and the rope, slings and anchor are chosen so the shear pin is always the weakest link. Searches take place on unstable ground; nothing here makes a dumpsite safe.

Assumptions

Table 1. Assumptions.

Quantity	Value	Basis
Bush friction, acetal on painted steel	0.15	Dry or dusty; low-friction acetal would be about 0.1
Rope friction on the drum	0.20	Polyester on painted steel, low side, so the tail force is not underestimated
Rope turns on the drum while hauling	4	Ship and rescue capstan practice
Shear pin material	S275 bar, ultimate shear 258 MPa (0.6 x 430 MPa)	Release spread taken as plus or minus 20 % over a batch
Design tension for every capstan part	8,000 N	Above the top of the pin's release band (7,648 N)
Sustained push per person on a bar	150 N	Short bouts; heaving can reach 300 to 400 N
Probe resistance, loosened debris	Cone 0.4 MPa, sleeve friction 1 kPa	No published data for dumpsite slides; a central case
Probe resistance, stiffer debris	Cone 1.0 MPa, sleeve friction 3 kPa	A compacted or wet case
Downward push on a probe	350 N one person, 650 N two	Sustained, leaning on the T handle
Subgrade modulus of slide debris	0.10 MN/m³	Very loose, low side
Searcher on a board	100 kg with a 1.5 dynamic factor	Kneeling and shifting weight
Timber battens	C16: 16 MPa bending, 8 GPa modulus	Treated softwood
Stretcher friction	0.5 on debris, 0.3 on boards	HDPE on mixed waste and on plywood
Sustained pull per hauler	200 N	Strap over the shoulder, 30 m
Carry limit per person	25 kg	100 m over rough ground
Walking speeds	2.5 m/s running empty, 0.8 m/s carrying	Rough ground
Opening the capstan crate and sharing out its load	60 s	Three people; estimate for the capstan wave from the host site

A. Hand capstan

The capstan meets R4 with four people, and its shear pin bounds the rope tension (R10).

- The rope runs at a 64.2 mm radius on the drum; friction in the foot bush adds the equivalent of 4.5 mm of lever per newton of rope tension [A1].
- A 5,000 N pull needs 343 N m at the head: 228 N at each bar end, so 114 N each for four people (two on each bar) or 228 N each for two, with hands 754 mm from the axis [A2].
- With four turns the tailer holds about 33 N on the tail [A3].

- The 6 mm S275 shear pin on a 60 mm radius releases at 7.29 kN, 438 N m, which is a rope tension of about 6,374 N, between 5,099 and 7,648 N over a batch [A4]. The lower bound stays above the 5 kN of R4.
- The 14 mm polyester rope (40 kN, 36 kN through the splice) has a factor of 7.2 at 5,000 N and 4.5 at the 8,000 N design tension [A5].
- The rope leaves the drum 15 degrees down to the hold-down roller. The post carries 0.70 kN m at 8,000 N: 63 MPa in the 60.3 x 5.0 tube against a 355 MPa yield [A6].
- One holding pawl alone takes 5.5 kN at the design tension: 77 MPa bearing on a 12 x 6 mm tooth face and 49 MPa shear in its 12 mm pin [A7]. The drive pawl takes 5.9 kN at pin release: 62 MPa on the 12 x 8 mm tooth face [A8].
- Each 40 x 40 x 2.5 bar carries 219 N m at its socket when the pin releases: 50 MPa [A9].
- Each quarter-turn stroke takes in about 101 mm of rope, about 1.5 m a minute at 15 strokes [A10].
- The rope leaves under the roller 38 mm above the ground and the anchor sling pulls from the eye at the same height, so there is no tipping moment; the stakes only stop the frame turning and skating [A11].
- The 10 mm anchor eye sees 42 MPa bearing from the 19 mm shackle pin at 8,000 N; the 2,000 kg round sling has a factor of 2.5 on its working load limit at the design tension [A12].

B. Probes

R2 is met on paper with the 18 mm tip on all six probes (decision 20A): one person reaches the full depth in loosened debris and two people in stiffer debris. The sleeve friction assumption must be confirmed on a test heap at TRL 4.

- Loosened debris: tip 102 N, friction 50 N per metre, 227 N to push to 2.5 m; one person reaches the full 2.6 m usable depth [B1c].
- Stiffer debris: tip 254 N, friction 151 N per metre, 631 N to push to 2.5 m. Two people reach the full 2.6 m; one person reaches 0.6 m, because a single person's push is used up by the tip and the friction [B1s].
- The 16 x 2.0 tube buckles at 4,558 N with 1 m free, 1,139 N with 2 m free and 506 N with 3 m free (pinned ends) [B2-1, B2-2, B2-3].
- The probe is 3,068 mm long assembled and weighs 3.07 kg; the 18 mm tip has a 3 mm rounded point; usable depth is about 2.6 m with the handle at chest height above the surface [B3].
- For comparison, the earlier 22 mm tip let one person reach nothing in stiffer debris and two people 1.8 m [B4].
- The tip now stands 1 mm proud of the 16 mm tube each side, against 3 mm before. The results assume the sleeve friction does not rise as that clearance falls; the test heap at TRL 4 must confirm it [B5].

C. Crawl boards

R3 is met on paper.

- A kneeling 100 kg searcher with a 1.5 dynamic factor, plus the 9.7 kg board, spread over 0.8 m of board gives 4.4 kPa and 44 mm of sinkage on very loose debris [C1].
- Bridging a 0.8 m void with 1,472 N at mid-span gives 294 N m; the battens alone see 4.0 MPa (factor 4.0 on C16) and deflect 0.8 mm [C2].
- The deck spans 300 mm clear between the battens; a 1,000 N knee on 150 mm of deck gives 13.3 MPa in 15 mm plywood [C3].
- Each board weighs 9.7 kg and each link plate 0.43 kg [C4].

D. Sheet stretcher

R5 is met under the drill-card rule of decision 21A: four haulers whenever the stretcher crosses debris, two only along the boards (153 N each). Two haulers on loose debris would need 255 N each, more than the 200 N sustained pull assumed, which is why the rule exists.

Table 2. Haul force for a 100 kg casualty on the 3.9 kg stretcher [D].

Surface	Total	Each of two haulers	Each of four haulers
Debris, level	510 N	255 N	127 N
Debris, 15 degrees up	756 N	378 N	189 N
Boards, level	306 N	153 N	76 N
Boards, 15 degrees up	559 N	280 N	140 N

E. Sheet clamp

Four M12 wing bolts at 2.5 kN each, with two faces at a friction coefficient of 0.3, hold about 6.0 kN before the sheet slips [E1]. That is below the shear pin's nominal release; a sheet that slips out of the clamp is safe.

F. Masses, carrying and deployment

Table 3. Carried items [F].

Item	Mass
Capstan frame with post and roller	20.9 kg
Capstan drum, head and ratchet	15.7 kg
Capstan bars (2)	5.2 kg
Stakes and mallet	9.2 kg
Rope (30 m)	4.1 kg
Slings, shackles and sheet clamp	7.6 kg
Crawl board (each of 4)	9.7 kg
Board link plates (8)	3.5 kg
Probes (6)	18.4 kg
Shovels (4)	9.2 kg
Sheet stretcher	3.9 kg
Lookout kit, lighting, PPE, bags and cards	13.5 kg

- The heaviest carried item is the capstan frame with its post and roller, 20.9 kg. The search crate (about 51 kg body and 17 kg lid as modelled in 18 mm boards) stays at the shed; the capstan crate (about 52 kg and 11 kg) stays at the host site [F1]. A reused crate's own mass will differ.
- The carried kit is 150 kg: 86 kg for the search wave and 64 kg for the capstan wave. Four people carrying everything at once would take 37.5 kg each [F2].
- Two waves (decision 22A): four people carry the search wave first, 21.5 kg each; a second group of three brings the capstan set, 21.4 kg each, under the 25 kg limit. R8 is met in two waves [F2b].
- Deployment from the alarm to the first probe line: 40 s to run 100 m to the shed, 30 s to open, 125 s to carry back, 20 s to post the lookout and 60 s to lay the first boards while the probes are joined: 275 s, 4.6 min [F3]. The capstan is not needed for the first probe line, so R1 is unchanged.
- The capstan set is kept at the host site (decision 23C). Allowing 60 s to open its crate and share out the load, the capstan wave arrives 3.8 min after the alarm from 100 m, 9.2 min from 300 m, 14.8 min from 500 m and 28.5 min from 1 km [F4]. Amish decided on 2026-10-04 (3A, HPL-DDR-004) that a shared capstan serves only sites within about 500 m walk of the host shed, where it arrives in about 14.8 min; the TRL 4 timed drills check this.

Decision 23C splits the kit into a search kit that every site keeps and one capstan set shared by neighbouring sites, puts each in a reused crate (or a bought steel job box where none can be had), and takes the PPE from the cooperative's own stock.

- The search kit with its crate costs USD 827.00 per site: crate and hardware USD 98.00, lighting USD 95.00, PPE USD 0 from the cooperative's stock (about USD 168 if bought) [G1].
- The shared capstan set (capstan, rope, slings, clamp and its own crate) costs USD 547.00 for each group of neighbouring sites: USD 273.50 a site if two share it, USD 182.33 if three [G1b].
- The search core (boards, links, probes, pins, shovels, stretcher, lookout kit and cards) is USD 526.00 [G2].

- R9 as restated (HPL-REQ-001 v0.4): USD 827.00 per site search kit against USD 850, USD 23.00 under [G2b]. A bought steel job box in place of the reused crate (about USD 300 to 600) would put the site over the restated target.
- Value-engineering target: USD 1,800. Estimated cost of the constructable design: USD 1,374.00 (USD 426.00 under the target), for the host site: its search kit and the shared capstan set [G3].

Results against the requirements

Table 4. Results (also written to `docs/04-calcs/results.csv`).

ID	Result	Status
R1	4.6 min estimated from alarm to first probe line	Met on paper
R2	18 mm tip: loosened debris 2.6 m by one person; stiffer debris 0.6 m for one, 2.6 m for two	Met on paper; confirm on a test heap at TRL 4
R3	44 mm sinkage; bridges a 0.8 m void at 4.0 MPa	Met on paper
R4	5 kN at 114 N each for four people; pin releases at 5.1 to 7.6 kN	Met on paper
R5	127 N each for four on level debris; 153 N each for two on the boards	Met with the drill-card rule: four haulers on debris, two only along the boards
R6	Stop rule printed on the lid and pocket cards	Not verifiable at TRL 3
R7	Weather-resistant materials; no lithium cells	Met on paper by material choice
R8	Heaviest item 20.9 kg; search wave 21.5 kg each for four, capstan wave 21.4 kg each for three	Met in two waves
R9	USD 827 per site search kit against USD 850; shared capstan set USD 547 per group of sites	Met as restated
R10	Pin releases at 6.4 kN nominal, 7.6 kN at most; rope factor 4.5 at 8 kN	Met on paper
R11	Two holding pawls; one takes the design tension	Met on paper

Requirement map (kit 1.9.1)

The requirements R1 to R11 are now PR-001 to PR-011 in HPL-PRS-001 (same targets). Results by product requirement: PR-001 deploy fast [F3]; PR-002 probe depth [B1c, B1s]; PR-003 crawl board [C1 to C3]; PR-004 capstan pull [A2 to A4]; PR-005 casualty transport [D]; PR-007 storage life (materials, BOM); PR-008 portable [F1 to F2b]; PR-009 cost [G1 to G3]; PR-010 overload limit [A4, A5]; PR-011 holds when let go [A7]. PR-012 lighting: six head lamps and two hand lamps on AA cells, BOM line 25, 1.5 kg [F-lighting]. PR-013 repair with common stock: steel tube and bar, plywood, timber, HDPE, rope (BOM). PR-014 not certified equipment: safety wording in the problem statement, README, drill card and drawing title blocks. PR-015 shared capstan arrival: 14.8 min from 500 m, 3.8 min from 100 m [F4]. The numbers for options B and C are in `docs/04-calcs/options_calc.py` (HPL-OPT-001).

Pin bar (kit 1.9.1, HPL-DDR-006): A4 uses 430 MPa nominal. EN 10025-2 allows 410 to 560 MPa for S275 bar of 3 to 100 mm, so the pin bar is bought with a mill certificate at 480 MPa or less and three pins per batch are release-tested; the certified range with 5 % test scatter gives a release of 5.77 to 7.47 kN (HPL-STM-001, S5 and S6).